

Graphing Basic Quadratic Equations

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A **quadratic relationship** is represented by an equation whose highest exponent of the variable is 2.

Parabola

The graph of a quadratic equation is called a **parabola**.

$$y = x^2$$

The graph is a parabola.

Vertex

The graph of $y = ax^2$ has its **vertex** at $(0, 0)$.

$$y = 2x^2$$

Vertex: $(0, 0)$

Direction of Opening

If $a > 0$, the parabola **opens upward**.

If $a < 0$, the parabola **opens downward**.

$$y = x^2$$

$$y = -x^2$$

Vertical Shift

In $y = ax^2 + b$, the graph is shifted **vertically** by b units.

$$y = x^2 + 4$$

The graph is shifted 4 units upward.

Horizontal Shift

In $y = a(x - h)^2$, the graph is shifted **horizontally**.

Its vertex is at $(h, 0)$.

$$y = (x - 3)^2$$

Vertex: $(3, 0)$

Width of the Parabola

The value of $|a|$ affects the **width** of the parabola.

Larger values make it narrower, while smaller positive values make it wider.

Graphing a Quadratic

To graph a quadratic equation, make a **table of values**, plot the ordered pairs, and draw a smooth curve.

$$y = x^2$$

$$(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)$$

Axis of Symmetry

The **axis of symmetry** passes through the vertex of the parabola.

For

$$y = (x - 3)^2$$

the vertex is $(3, 0)$.

The axis of symmetry is

$$x = 3.$$

Examples

Example 1

Complete a table of values and graph

$$y = x^2$$

using $x = -3, -2, -1, 0, 1, 2, 3$.

Example 2

Complete a table of values and graph

$$y = -2x^2$$

using $x = -2, -1, 0, 1, 2$.

Example 3

Complete a table of values and graph

$$y = \frac{1}{2}x^2$$

using $x = -4, -2, 0, 2, 4$.

Example 4

Complete a table of values and graph

$$y = x^2 + 4$$

Identify the vertex and state how the graph differs from $y = x^2$.

Example 5

Complete a table of values and graph

$$y = -x^2 - 2$$

Identify the vertex and describe the direction of opening.

Example 6

Complete a table of values and graph

$$y = (x - 3)^2$$

Identify the vertex and axis of symmetry.

Example 7

Complete a table of values and graph

$$y = -2(x + 1)^2$$

Identify the vertex and direction of opening.

Practice Exercises

Practice 1

Complete a table of values and graph

$$y = x^2$$

using $x = -3, -2, -1, 0, 1, 2, 3$.

Practice 2

Complete a table of values and graph

$$y = -2x^2$$

using $x = -2, -1, 0, 1, 2$.

Practice 3

Complete a table of values and graph

$$y = \frac{1}{2}x^2$$

using $x = -4, -2, 0, 2, 4$.

Practice 4

Complete a table of values and graph

$$y = x^2 + 4$$

Identify the vertex and state how the graph differs from $y = x^2$.

Practice 5

Complete a table of values and graph

$$y = -x^2 - 2$$

Identify the vertex and describe the direction of opening.

Practice 6

Complete a table of values and graph

$$y = (x - 3)^2$$

Identify the vertex and axis of symmetry.

Practice 7

Complete a table of values and graph

$$y = -2(x + 1)^2$$

Identify the vertex and direction of opening.

Activities

Activity 1

Complete a table of values and graph

$$y = 2x^2$$

using $x = -3, -2, -1, 0, 1, 2, 3$.

Activity 2

Complete a table of values and graph

$$y = -x^2$$

using $x = -2, -1, 0, 1, 2$.

Activity 3

Complete a table of values and graph

$$y = \frac{3}{4}x^2$$

using $x = -4, -2, 0, 2, 4$.

Activity 4

Complete a table of values and graph

$$y = x^2 - 3$$

Identify the vertex and state how the graph differs from $y = x^2$.

Activity 5

Complete a table of values and graph

$$y = -x^2 + 5$$

Identify the vertex and describe the direction of opening.

Activity 6

Complete a table of values and graph

$$y = (x + 2)^2$$

Identify the vertex and axis of symmetry.

Thank you!